

# Containerlab Node Manager — Student Quick Start

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This guide walks you through the manager from the browser: deploying a lab, working on its devices from the command line, and saving your progress to Git so nothing is lost even after the lab itself is destroyed. Pick the path that matches what you were given and follow it step by step; each step says what to do and what you should see.

| Your starting point                           | Follow     |
|-----------------------------------------------|------------|
| I was given a lab by my instructor            | Scenario A |
| I want to build my own lab and keep it in Git | Scenario B |

|                                                         |    |
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## Before you start

| What you have been given |                                                                                                                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Manager address          | <code>http://192.168.132.132:8081</code> — open it in a browser; there is no sign-in                                                                                         |
| Instructor materials     | The <code>link-basics</code> lab is already staged on the lab VM, with its starting point, finished version and a broken version saved in the course repository (Scenario A) |
| Course repository        | <code>https://github.com/pruger-dev/netlab-course</code> — the instructor's copy; you read it, you never push to it                                                          |

|                     |                                                                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Your own repository | Scenario A: your own copy of the course repository, made from its template before you begin. Scenario B: a brand-new, empty repository you create in step B5 |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|

Everything below happens in one of three places: **your computer** (the browser you are reading this in, and any terminal you run on your own machine), **the lab VM** (the server that actually runs the lab's devices, and where the manager itself lives), and **your online repository** (your account on GitHub, out on the internet). A file can exist in one of these and not the others until you deliberately move it — keeping track of which place you are looking at is most of what this guide teaches.

A few Git words come up along the way. A **repository** is a project's saved history, kept on GitHub and also as a folder copy elsewhere; a **folder** inside it groups related files, the same as a folder on your computer; a **commit** is one saved snapshot of some files with a short message; to **push** — this guide also calls it **upload** — is to send your commits to the copy on GitHub; and to **clone** is to copy a whole repository from GitHub onto another machine. Each word is used again later without repeating its definition.

## Scenario A: use a lab your instructor gave you

In this scenario you deploy a lab your instructor already prepared, look at it from the command line, make a small change, save your work to Git, and practice recovering from a fault — all using `link-basics`, two routers, `r1` and `r2`, joined by one link.

**Save progress saves device configurations, their restore artifacts and a manifest into your repository folder. It does not save the topology file, the map or a README.** For `link-basics` the topology file is already staged on the lab VM and included in the course repository, so you never need to save it yourself in this scenario.

### A1 Open the manager

**Action** In a browser on your computer, go to `http://192.168.132.132:8081`.

**Expected result** Home shows a **Deploy** box, a **Build** box, a **Manager** ▾ menu and a lab list. Before anything is deployed the list reads **No labs yet**.

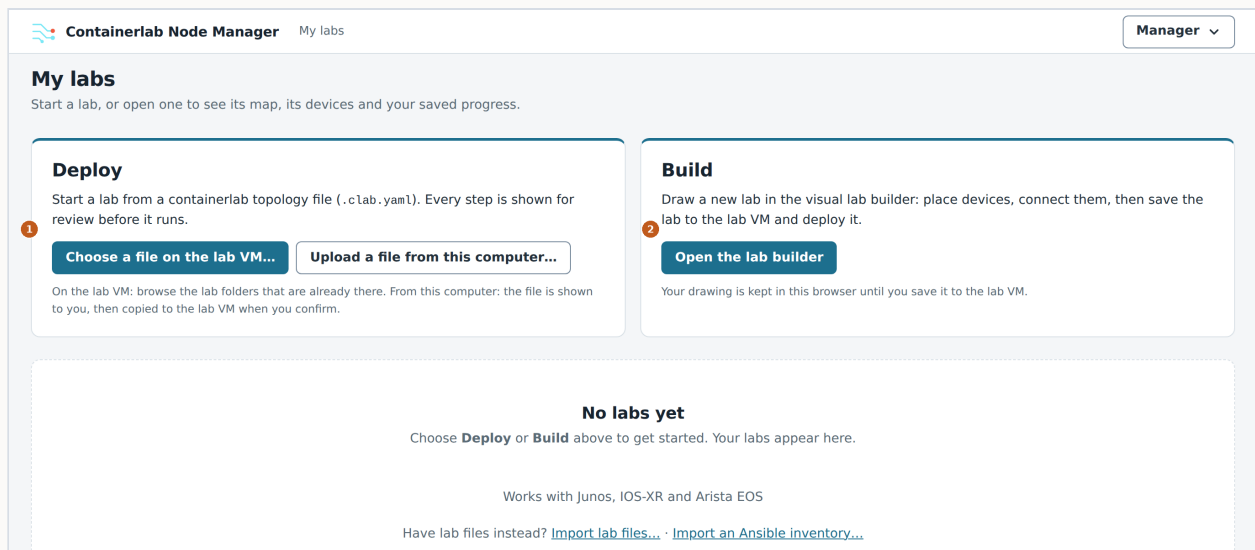


Figure A.1 — Home page before any lab is deployed.

### A2 Deploy the lab

**Action** In the Deploy box, click **Choose a file on the lab VM...** The **Deploy a new lab** dialog opens on the lab folders on the VM; open the `link-basics` folder and click ◇ **link-basics.clab.yml**.

**Expected result** A **Topology file** dialog opens showing the file location on the VM and a read-only preview of the topology (YAML), with buttons **Preview topology**, **Edit visually...**, **Add to My labs without starting** and **Deploy lab**.

**Action Click Deploy lab.**

**Expected result** A review titled **Start link-basics?** opens: "Creates and starts the devices in this topology. Nothing is deleted; device logins open as the devices boot." Confirm with **Start lab**.

**If not** If `link-basics` does not appear in the folder browser, ask your instructor to confirm it is staged on the lab VM — the browser only shows lab folders the manager trusts.

Topology file

File location on the VM

/srv/containerlab-node-manager/projects/link-basics/link-basics.clab.yml

Topology (YAML)

```
# link-basics: two Arista cEOS routers joined by one link.
# Instructor lab package for the Containerlab Node Manager student quick start.
#
# r1 eth1 ----- eth1 r2
# 10.0.0.1/30      10.0.0.2/30  (addresses come from the saved state "reference/start")
#
# Keep `name:` and the node names exactly as they are: every saved configuration state in the
# course repository is recorded against clab-link-basics-r1 and clab-link-basics-r2. The
# management addresses are pinned so that a saved state applies after a redeploy as well.
# Logins are the kind's published default (admin / admin); nothing secret is kept here.
name: link-basics
mgmt:
  network: clab
  ipv4-subnet: 172.20.20.0/24
topology:
  nodes:
    r1:
      kind: arista_ceos
```

Deploy lab adds this lab to My labs and starts its devices on the VM. Add without starting keeps it in My labs only. Existing files can't be edited as text here — use Edit visually, or edit them on the VM.

Preview topology

Edit visually...

Add to My labs without starting

Deploy lab

Figure A.2a — The Topology file dialog for `link-basics.clab.yml`.

Start link-basics?

link-basics

/srv/containerlab-node-manager/projects/link-basics/link-basics.clab.yml

Runs this trusted topology with host privileges, including its configured hooks, mounts and image pulls.

Creates and starts the devices in this topology. Nothing is deleted; device logins open as the devices boot.

► Technical details

Runs on the lab VM. If the lab changes before you confirm, this check is repeated.

Cancel

Start lab

Figure A.2b — The Start link-basics? review.

**A3 Wait for the devices to become ready**

**Action** Stay on the lab page while `r1` and `r2` start.

**Expected result** A running container is not yet a router that accepts a login. Right after you confirm, the header reads **Starting lab** and **0 of 2 devices ready**, and each device's row reads **Unavailable** for a moment, then **Starting**, then **Ready** once the manager can log in and get an answer from it. This usually takes one to two minutes; on a lightly loaded VM it can take under a minute. Wait until the header reads **2 of 2 devices ready**.

**If not** A pill reading **Needs credentials** means the device is up but the manager could not log in — check the login with your instructor. **Needs attention** after several failed attempts offers **Test login now** to retry once the problem is fixed.

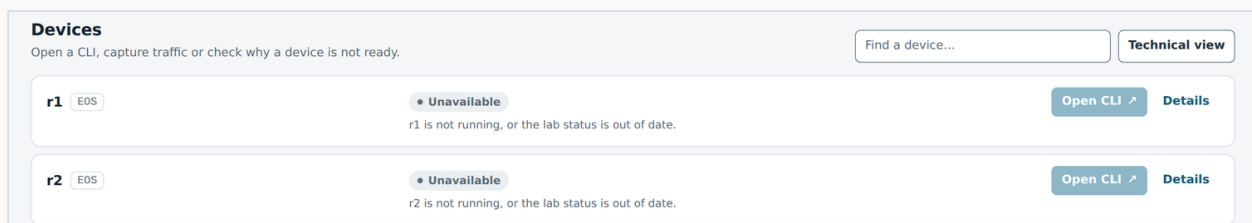


Figure A.3a — Just after confirming: 0 of 2 devices ready.

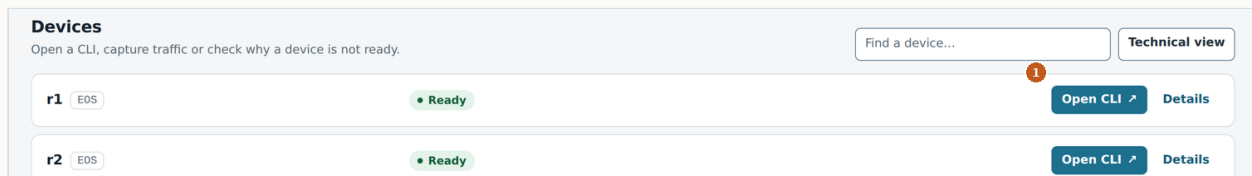


Figure A.3b — Both devices read Ready.

**A4 Look around from the command line**

**Action** On the Devices tab, click **Open CLI** next to `r1`. It opens a terminal in a new browser tab; wait until its status reads **Connected**, then at the `r1>` prompt run:

```
r1>show version
r1>show interfaces status
r1>show ip interface brief
```

**Expected result** `show version` reports the platform; `show interfaces status` shows `Et1` as `connected`, since the link between `r1` and `r2` is physically up; `show ip interface brief` lists only `Management0` — `Ethernet1` has no address yet, so EOS does not list it there. That is expected: the lab starts with the link wired but nothing configured on it.

```

Last login: Tue Sep 22 17:27:27 2026 from 172.20.20.1
r1>show version
Arista cEOSLab
Hardware version:
Serial number: 4DD62532A07EB6D99D4C174D07BC5FF1
Hardware MAC address: 001c.73ec.f806
System MAC address: 001c.73ec.f806

Software image version: 4.35.0F-44178984.4350F (engineering build)
Architecture: i686
Internal build version: 4.35.0F-44178984.4350F
Internal build ID: 8d2eca31-2c2b-42e4-9429-bc70c71397c8
Image format version: 1.0
Image optimization: None

Kernel version: 6.8.0-139-generic

Uptime: 0 minutes
Total memory: 76299668 kB
Free memory: 64854652 kB

r1>show interfaces status
Port      Name      Status      Vlan    Duplex Speed  Type      Flags Encapsulation
Et1       connected
Ma0       connected  routed      a-full  a-1G   10/100/1000
r1>

```

Figure A.4 — r1's terminal after the three show commands.

## A5 Connect your repository

**Action** Open the **Progress** tab. With nothing connected yet it shows **Choose where to save your progress**; click **Connect a repository by URL**. Paste `https://github.com/pruger-dev/netlab-course-student.git`, set the folder to `link-basics/work`, tick the acknowledgement that complete device configurations will be saved and uploaded, and click **Connect repository**.

**Expected result** The destination line reads exactly "link-basics saves to netlab-course-student > link-basics/work > latest/". **Saved versions** now also shows a group, **Instructor and reference versions** — "Versions your instructor put in the repository appear here. Apply one to load it onto your running devices; the current configuration is backed up first." — listing three rows, each with 5 files and **View**, **Compare with my latest save** and **Apply to running lab...: Starting state** ( `link-basics/reference/start/latest` ), **Final state (instructor)** ( `link-basics/reference/solution/latest` ) and **Troubleshooting scenario 01** ( `link-basics/reference/broken-01/latest` ).

**If not** If the manager reports it cannot reach or push to the repository, check that you made your own copy of the course repository with **Use this template > Create a new repository** on GitHub first — the URL above must point at your copy, not the instructor's.

**Connect a repository by URL** ✕

Paste the repository's HTTPS address from GitHub (Code > HTTPS). The lab VM's existing GitHub login is used — you are never asked for a password or token.

1 **Repository URL**

2 **Folder for this lab** optional

3 ☒ I understand that complete device configurations — including any passwords or keys they contain — will be saved to this repository and uploaded to its online copy.

4 **Cancel** **Connect repository**

Figure A.5a — The Connect a repository by URL dialog, filled in.

1 **Saving to netlab-course-student > link-basics/work**

Not saved yet · Save progress creates a snapshot you can return to later.

**Save progress** **Create checkpoint...** **More** ▾

**Saved versions**

Versions of this lab you can return to. Save progress creates a new one.

**Browse the repository...** **Full history...**

**Latest**

You haven't saved this lab yet. Save progress creates a configuration snapshot you can return to later.

**Checkpoints**

No checkpoints yet. Create a checkpoint when you reach an important milestone.

**Instructor and reference versions**

Versions your instructor put in the repository appear here. Apply one to load it onto your running devices; the current configuration is backed up first.

|                                                                              |         |             |                                    |                                |
|------------------------------------------------------------------------------|---------|-------------|------------------------------------|--------------------------------|
| <b>Troubleshooting scenario 01</b><br>Link-basics/reference/broken-01/latest | 5 files | <b>View</b> | <b>Compare with my latest save</b> | <b>Apply to running lab...</b> |
| <b>Final state (instructor)</b><br>Link-basics/reference/solution/latest     | 5 files | <b>View</b> | <b>Compare with my latest save</b> | <b>Apply to running lab...</b> |
| <b>Starting state</b><br>Link-basics/reference/start/latest                  | 5 files | <b>View</b> | <b>Compare with my latest save</b> | <b>Apply to running lab...</b> |

2

Figure A.5b — The Progress tab once the repository is connected.

## A6 Load the starting configuration

**Action** On the Progress tab, find **Starting state** and click its **Apply to running lab...**

**Expected result** A review titled **Replace running configuration** opens, its source line reading "Source: Starting state · netlab-course-student > link-basics/reference/start/latest · saved ... · <commit>". Each device is listed, for example "r1 EOS 6 differences from the running configuration". Tick "I understand the running configuration on the selected devices will be replaced." and click **Replace configurations**.

**Expected result** Within about a dozen seconds each device reports "Configuration replaced and verified against the saved desired state." Go back to r1's terminal tab from A4 (still open) and run `ping 10.0.0.2` — it succeeds, since `start` addresses the link `10.0.0.1/30` on r1 and `10.0.0.2/30` on r2.

## Replace running configuration

Lab: **link-basics**

1 Source: **Starting state** · **netlab-course-student** > **link-basics/reference/start/latest** · saved 35 minutes ago · 089a35aeb8

Current configurations are backed up first. The devices are not rebooted.

Devices

- ☒ **r1** EOS 6 differences from the running configuration
- ☒ **r2** EOS 6 differences from the running configuration

- Each device's current configuration is backed up first. A device whose backup fails is left unchanged.
- The devices are not rebooted. The device itself checks the new configuration when it is activated, and the change is undone on its own if the device cannot be reached again within 5 minutes.
- Devices that are skipped above are not touched.

► **Advanced options**

2 ☐ I understand the running configuration on the selected devices will be replaced.

3

Cancel Replace configurations

Figure A.6a — Replace running configuration, reviewing Starting state.

## Configuration replaced

• Configuration replaced

Configuration replaced on 2 devices.

• Replaced and verified **r1** EOS

► Details

• Replaced and verified **r2** EOS

► Details

► Details

Figure A.6b — Both devices replaced and verified.

### A7 Make a change on r1

**Action** On **r1**'s CLI:

```
r1>enable
r1#configure terminal
r1(config)#interface Loopback0
r1(config-if-Lo0)#description r1 loopback
r1(config-if-Lo0)#ip address 10.255.0.1/32
r1(config-if-Lo0)#end
r1#write memory
r1#show ip interface brief
```

On EOS a configuration command is active the moment you enter it; **write memory** only keeps it across a device restart, and **Save progress** in the next step captures the running configuration either way.

**Expected result** `show ip interface brief` now also lists `Loopback0` with address `10.255.0.1/32`. `r2` is untouched for now — its own loopback comes in step A9.

```
Last login: Tue Sep 22 17:28:10 2026 from 172.20.20.1
r1>enable
r1#configure terminal
r1(config)#interface Loopback0
r1(config-if-Lo0)#description r1 loopback
r1(config-if-Lo0)#ip address 10.255.0.1/32
r1(config-if-Lo0)#end
r1#write memory
Copy completed successfully.
r1#show ip interface brief
```

| Interface   | IP Address      | Status | Protocol | MTU   | Address Owner |
|-------------|-----------------|--------|----------|-------|---------------|
| Ethernet1   | 10.0.0.1/30     | up     | up       | 1500  |               |
| Loopback0   | 10.255.0.1/32   | up     | up       | 65535 |               |
| Management0 | 172.20.20.11/24 | up     | up       | 1500  |               |

```
r1#
```

Figure A.7 — `Loopback0` added and read back on `r1`.

## A8 Save your progress

**Action** Click **Save progress** in the lab header.

**Expected result** The manager captures both devices and opens **Review before uploading**: "What this save changed compared with the previous one. Configuration files may contain passwords or keys." and "This save is on the lab VM only. Nothing is uploaded to github.com unless you choose Upload these changes." Because this is the first save, every file is listed as added: `r1.cfg` added, `r1.eoscfg` added, `r2.cfg` added, `r2.eoscfg` added. **Upload these changes** sends it to your repository at once; **Not now — keep it on the VM** leaves it waiting under **Recent saves** instead. Click **Upload these changes**.

**Expected result** The Progress tab reads **Saved to Git just now**, and the **Latest** row reads "netlab-course-student > link-basics/work > latest · Saved just now · 5 files". On GitHub, `link-basics/work/latest/` now holds `manifest.json`, `r1.cfg`, `r2.cfg`, `r1.eoscfg` and `r2.eoscfg`.

Review before uploading

What this save changed compared with the previous one. Configuration files may contain passwords or keys.

This save is on the lab VM only. Nothing is uploaded to github.com unless you choose **Upload these changes**.

Details

r1.cfg • added

r1.eoscfg • added

r2.cfg • added

r2.eoscfg • added

Open the full saved version

Not now — keep it on the VM

Upload these changes

Figure A.8a — Review before uploading, everything added.

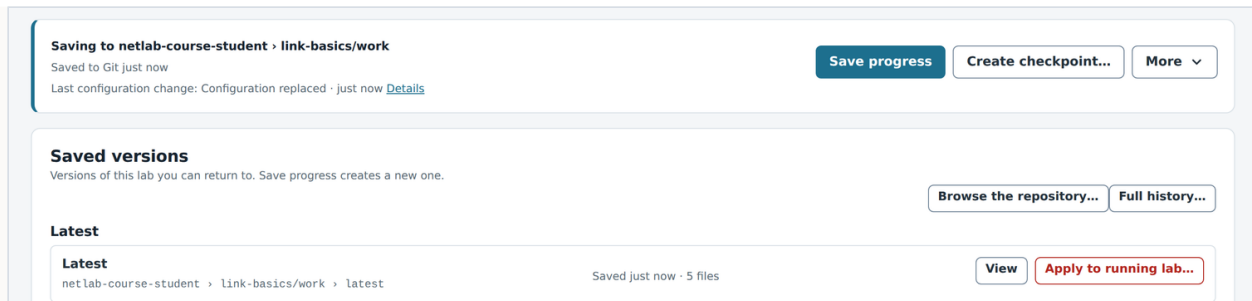


Figure A.8b — Saved to Git just now.

## A9 Checkpoint, then add r2's loopback

**Action** Open the **Save progress** menu (▼) and choose **Create checkpoint....** Name it `loopback-added` and confirm; it opens the same **Review before uploading** — click **Upload these changes**.

**Action** Now give `r2` its own loopback: on `r2`'s CLI, run the same commands as in A7, using `description r2 loopback` and `ip address 10.255.0.2/32`, then `write memory`. Click **Save progress** again and upload that review too.

**Expected result** This second save updates the same `link-basics/work/latest` in place — it never creates a nested `latest/latest`. **Saved versions** now lists **Latest** ("Saved just now · 5 files") and, under **Checkpoints**, `loopback-added`. On GitHub, `link-basics/work/latest/r2.cfg` now contains `Loopback0`, while `link-basics/work/checkpoints/loopback-added/r2.cfg` does not — the checkpoint captured `r2` before its loopback existed.

## Create checkpoint ✕

Saving to netlab-course-student > link-basics/work

A checkpoint is a named version you can return to later.

**Checkpoint name**

loopback-added

Letters, numbers, underscores or hyphens. Each name can be used once.

Saved as: loopback-added

**What did you get working? (optional)**

For example: OSPF adjacencies up on every router

☒ Upload after saving — you see what changed first and confirm the upload

► **Devices removed from this lab**

Cancel
Create checkpoint

Figure A.9a — Naming the checkpoint `loopback-added`.

### Saved versions

Versions of this lab you can return to. Save progress creates a new one.

Browse the repository... Full history...

**Latest**

**Latest**  
netlab-course-student > link-basics/work > latest

Saved just now · 5 files

View
Apply to running lab...

**Checkpoints**

**loopback-added**

Saved just now · 5 files

View
Compare with my latest save
Apply to running lab...

**Instructor and reference versions**

Versions your instructor put in the repository appear here. Apply one to load it onto your running devices; the current configuration is backed up first.

**Troubleshooting scenario 01**  
link-basics/reference/broken-01/latest

5 files

View
Compare with my latest save
Apply to running lab...

**Final state (instructor)**  
link-basics/reference/solution/latest

5 files

View
Compare with my latest save
Apply to running lab...

**Starting state**  
link-basics/reference/start/latest

5 files

View
Compare with my latest save
Apply to running lab...

Figure A.9b — Saved versions listing Latest and the new checkpoint.

## A10 Break it, then recover

**Action** Find **Troubleshooting scenario 01** and click its **Apply to running lab...**, then confirm with **Replace configurations**.

**Expected result** Both devices are replaced. On `r1`'s CLI, `ping 10.0.0.2` now fails, and `show interfaces status` shows `Et1` as `notconnect` — the fault is on `r2` (its `Ethernet1` is administratively shut down), so `r1` sees the link go down too.

**Action** Recover by applying your own saved progress: find **Latest** under **Saved versions** and click its **Apply to running lab...**, then **Replace configurations** again. Its source line names your own folder: "Source: work · netlab-course-student > link-basics/work/latest · ...".

**Expected result** `ping 10.0.0.2` from `r1` succeeds again and both loopbacks are back. The *Save location* card still shows the lab saving to `link-basics/work` — applying a saved version never changes where the lab saves. Curious what changed since your checkpoint? The `loopback-added` row's **Compare with my latest save** opens a dialog titled **Compared with your latest save**.

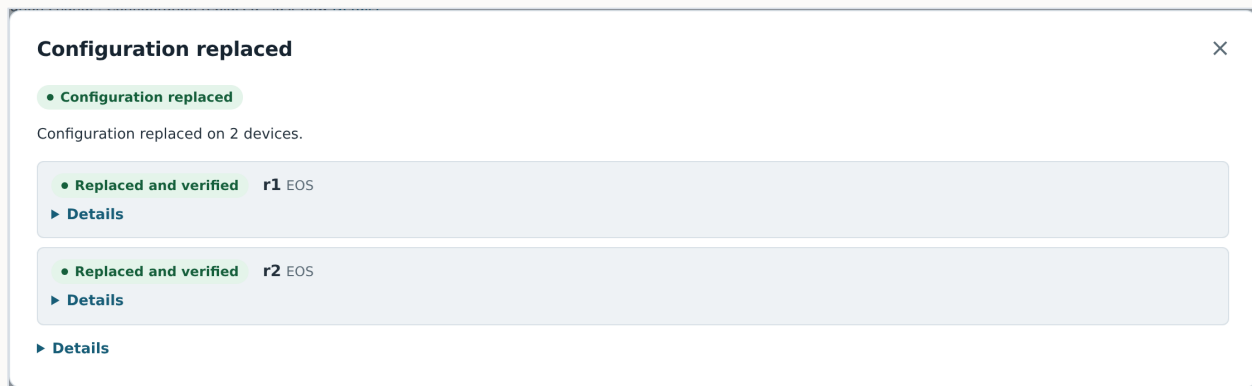


Figure A.10a — Troubleshooting scenario 01 applied; the link is down.

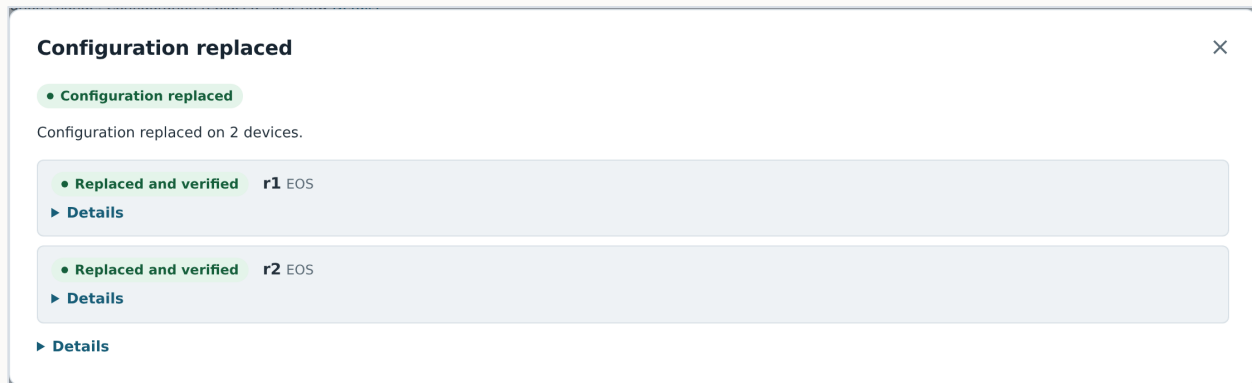


Figure A.10b — Your own Latest applied; the link is restored.

### **A11** Next time you open this lab

**Action** Close the browser, then open the manager again at `http://192.168.132.132:8081`.

**Expected result** The `link-basics` card on Home reads "link-basics · Running · 2 of 2 devices ready · Last saved N minutes ago · Deployed N minutes ago", with an **Open lab**

button. Open it — the Progress tab still lists **Latest** and your loopback-added checkpoint, exactly as you left them.

**Action** Open **Lab actions** ▼ to see everything you can do with this lab: **Start stopped devices** (or **Stop devices** and **Restart devices** while it runs), **Sync topology from VM**, **Packet capture...**, **Lab files...**, **All lab operations...**, **Redeploy lab...**, **Redeploy and clear the lab folder...**, **Destroy lab...**, **Remove from this manager...**, and **Advanced options** folding out **Import map...**, **Edit map**, **Telemetry settings...** and **Operation history...**. Open **Destroy lab...** to see its review — "The running devices are removed from the VM. Configuration changes you have not saved are lost. Your saved progress, checkpoints and backups remain." with a **Save progress first** option — then click **Cancel**; there is no need to destroy this lab now. **Remove from this manager...** is even less drastic: "Nothing on the lab VM changes: the running devices and the topology files stay. Progress you saved to Git stays in the repository."

## link-basics



● **Running**

2 of 2 devices ready

Last saved 1 minute ago

Deployed 3 minutes ago

**Open lab**

Figure A.11a — The lab card on Home, next time you open the manager.

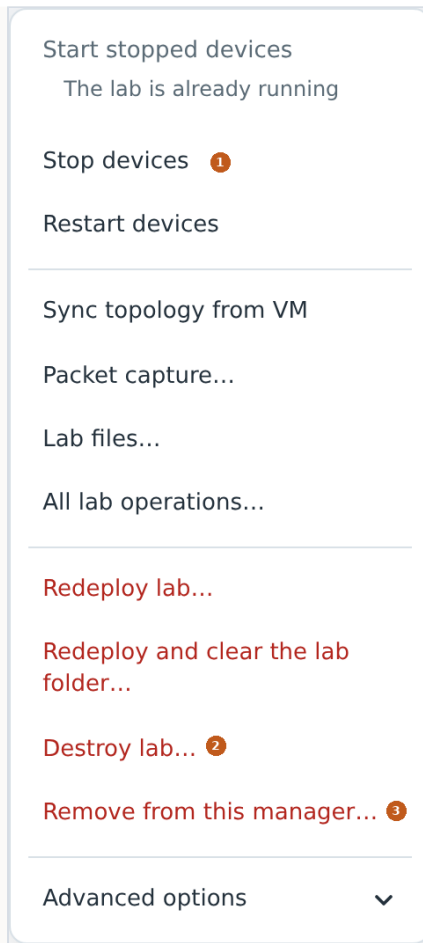


Figure A.11b — The Lab actions ▼ menu.

## Checklist

- [ ] `link-basics` deployed and both devices read **Ready**
- [ ] Repository connected, saving to `link-basics/work`
- [ ] **Starting state** applied and the link pings
- [ ] A loopback added on `r1`, checkpointed, then a loopback added on `r2`
- [ ] Progress **Saved to Git just now**
- [ ] Checkpoint `loopback-added` created
- [ ] **Troubleshooting scenario 01** applied, then your own **Latest** applied to recover

## Scenario B: build your own lab and keep it in Git

In this scenario you draw your own two-router lab in the visual builder, deploy it, configure it, and set up your own repository so your work survives even after the lab is destroyed and removed from the manager. The devices are `r1` and `r2`, joined by one link, both Arista cEOS.

**Save progress saves device configurations, their restore artifacts and a manifest into your repository folder. It does not save the topology file, the map or a README.** A lab definition and its saved device configurations are two different things — this scenario ends with both of them safely in Git, and step B7 is exactly the extra step that gets the topology and map there.

### B1 Start a new lab in the builder

**Action** From Home, click **Open the lab builder** in the Build box. The builder shows **Build a lab** with **New lab...** and **Open a draft...**; click **New lab...**. Fill in **Lab name** (`my-first-lab` — letters, digits, dot, dash and underscore only), **Start from** (**Two devices, one link**), **Device type for the starter** (**Arista cEOS · n24l/ceos:4.35.0F** — the image shown is only a suggestion, taken from labs already in My labs or a default) and **Lab folder on the VM** (`/srv/containerlab-node-manager/projects`). Click **Create draft**.

**Expected result** The editor opens with two devices, named `ceos1` and `ceos2` by default, already joined by one link. The status pill reads **Draft · kept in this browser only · not on the VM yet**.

Figure B.1 — The New lab dialog, filled in.

**B2 Draw and check the topology**

**Action** Right-click the first device and choose **Edit Node**. In the Node Editor panel (tabs Basic / Configuration / Runtime / Network / Advanced), set **Node Name** to `r1` and, on the Basic tab, **Image** `n24l/ceos` and **Version** `4.35.0F`; on the Network tab, set **Management IPv4** to `172.20.20.21`. Click **Apply**. Repeat for the second device: name it `r2`, the same image and version, and Management IPv4 `172.20.20.22`. A saved management address like this one is recommended because it applies again after a redeploy.

Changing something on the Network tab can reset the image back to its suggested name, so glance at the Basic tab again before you click **Apply**, and confirm with **View YAML** that each device still reads `image: n24l/ceos:4.35.0F`. While you are there, right-click empty canvas and choose **Add Text** to note what the lab is, for example "My first lab: r1 eth1 - r2 eth1".

**Expected result View YAML** shows the topology: `name: my-first-lab`, both nodes with `kind: arista_ceos`, `image: n24l/ceos:4.35.0F` and their management addresses, and one link, `r1:eth1 - r2:eth1`.

The screenshot shows the 'Node Editor' window with the 'EDIT' tab selected. The 'BASIC' sub-tab is active, displaying the following configuration for node 'r1':

- Node Name:** r1
- Kind:** arista\_ceos
- Image:** n24l/ceos
- Version:** 4.35.0F
- Icon:** Switch (with an 'EDIT' button)
- Label & Direction:**
  - Label Position: Bottom
  - Label Text Direction: Horizontal
  - Label Background: #000000
  - Transparent: ☐

An 'APPLY' button is located at the bottom right of the editor.

Figure B.2a — The Node Editor, setting r1's name, image and version.

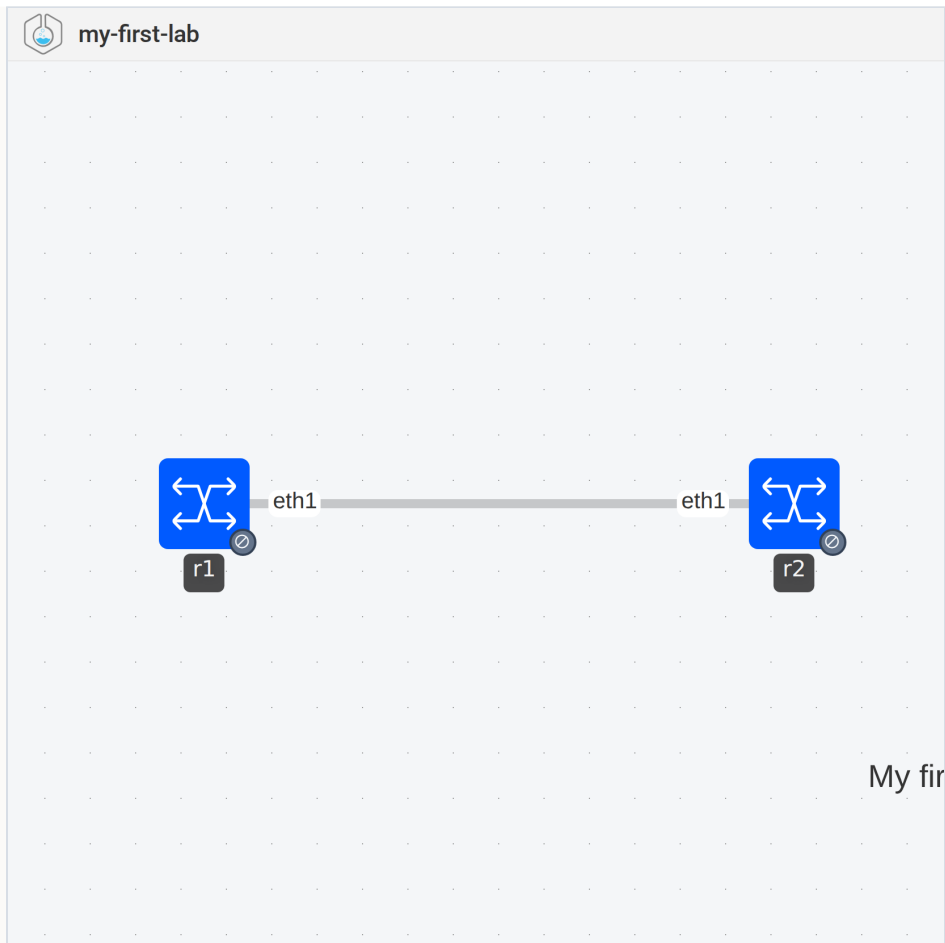


Figure B.2b — Finished canvas: r1, r2 and the note.

### B3 Save to the VM and deploy

**Action** Click **Save to the VM...** A review titled **Save my-first-lab to the VM?** shows the lab folder, `/srv/containerlab-node-manager/projects/my-first-lab/my-first-lab.clab.yml`, and the YAML that will be written; confirm with **Save lab**.

**Expected result** The result reads "✓ Save lab to the VM succeeded" and the status pill changes to **Saved on the VM**.

**Action** Click **Deploy or add this lab...**, then in the **Topology file** dialog click **Deploy lab**. A review titled **Start my-first-lab?** opens; confirm with **Start lab**.

**Expected result** The builder page itself does not navigate anywhere next — click ← **My labs** at the top left and open the `my-first-lab` card from Home to continue. On the lab VM, the project folder now holds `my-first-lab.clab.yml`, `my-first-lab.clab.yml.annotations.json` and the generated `clab-my-first-lab` folder.

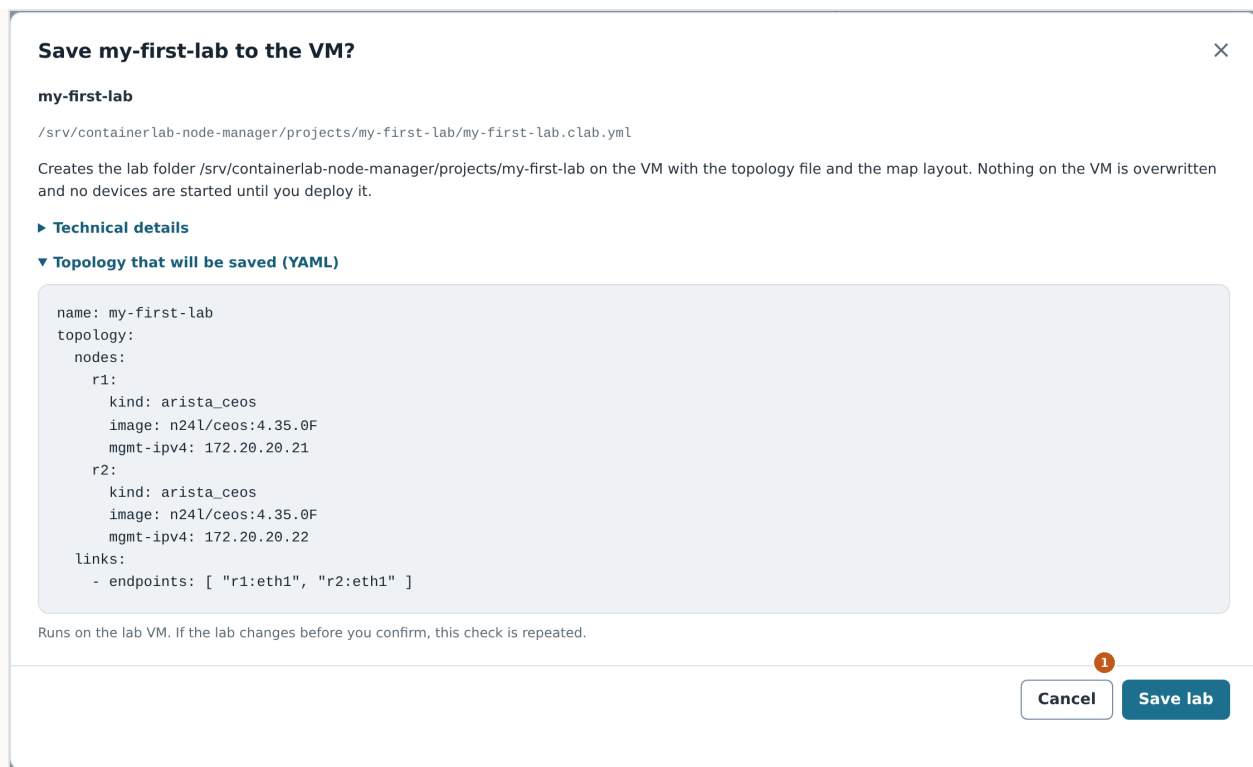


Figure B.3 — Save my-first-lab to the VM?, reviewing the YAML.

#### B4 Bring the link up

**Action** Wait for both devices to read **Ready** — around 40 seconds on this VM. Open a CLI on `r1` and run:

```
r1>enable
r1#configure terminal
r1(config)#ip routing
r1(config)#interface Ethernet1
r1(config-if-Et1)#no switchport
r1(config-if-Et1)#ip address 10.0.0.1/30
r1(config-if-Et1)#end
r1#write memory
```

`no switchport` matters here: without it, EOS keeps `Ethernet1` as a switched port and refuses the `ip address` that follows. On `r2`, run the same commands with `ip address 10.0.0.2/30`. From `r1`, run `ping 10.0.0.2`.

**Expected result** The ping succeeds.

```

Last login: Tue Sep 22 18:43:07 2026 from 172.20.20.1
r1>ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 72(100) bytes of data:
80 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.132 ms
80 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.007 ms
80 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.003 ms
80 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.005 ms
80 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.005 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.003/0.030/0.132/0.050 ms, ipg/ewma 0.053/0.079 ms
r1>

```

Figure B.4 — Both devices Ready, link addressed and pinging.

## B5 Create your own repository on GitHub

**Where to run it** On your computer.

**Action** On GitHub, choose **New repository**. Name it `my-network-labs`, set it **Private**, tick **Add a README file**, and click **Create repository**. From a terminal with GitHub's command-line tool installed, the same repository can be created with:

```
$ gh repo create my-network-labs --private --add-readme
```

Copy its **Code** → **HTTPS** address for the next step.

**Expected result** The repository exists, private, on branch `main`, with a `README.md`.

## B6 Connect the repository and save your progress

**Action** In the manager, open `my-first-lab` → **Progress** tab → **Connect a repository by URL**. Paste `https://github.com/pruger-dev/my-network-labs.git`, set the folder to `my-first-lab`, tick the acknowledgement that complete device configurations will be saved and uploaded, and click **Connect repository**.

**Expected result** The destination line reads exactly "my-first-lab saves to my-network-labs > my-first-lab > latest/".

**Action** Click **Save progress. Review before uploading** lists everything as added — `r1.cfg` added, `r1.eoscfg` added, `r2.cfg` added, `r2.eoscfg` added, since this is the first save — then click **Upload these changes**.

**Expected result** The status reads **Saved to Git just now**. On GitHub, `my-first-lab/latest/` holds `manifest.json`, `r1.cfg`, `r2.cfg`, `r1.eoscfg` and `r2.eoscfg` — and no topology file: that still only exists on the lab VM until the next step.

**Connect a repository by URL** ✕

Paste the repository's HTTPS address from GitHub (Code > HTTPS). The lab VM's existing GitHub login is used — you are never asked for a password or token.

- Repository URL**
- Folder for this lab** optional

One repository can hold several labs, each in its own folder. Clear it to save at the top level of a single-lab repository.

- ☒ I understand that complete device configurations — including any passwords or keys they contain — will be saved to this repository and uploaded to its online copy.  
[Details](#)

Cancel Connect repository

Figure B.6a — Connecting `my-network-labs` with folder `my-first-lab`.

**Saving to my-network-labs > my-first-lab**  
 Saved to Git just now Save progress Create checkpoint... More ▾

**Saved versions**  
 Versions of this lab you can return to. Save progress creates a new one. Browse the repository... Full history...

**Latest**  
 my-network-labs > my-first-lab > latest Saved just now · 5 files View Apply to running lab...

Figure B.6b — The first `Save progress`, uploaded.

## B7 Publish the topology files

**Where to run it** On the lab VM, as your ordinary account.

**Action** `Save progress` only uploads device configurations — the topology file and its map are still only on the lab VM. Publish them into the same checkout the manager just used:

```
$ cd ~/labs/my-network-labs
$ mkdir -p my-first-lab
$ cp /srv/containerlab-node-manager/projects/my-first-lab/my-first-lab.clab.yml*
  my-first-lab/
```

Write `my-first-lab/README.md`, for example:

```
# my-first-lab
Two Arista cEOS routers, r1 and r2, joined by one link (r1 eth1 - r2 eth1).
Link addresses: 10.0.0.1/30 on r1, 10.0.0.2/30 on r2.
Saved progress lives in latest/ (and checkpoints/ for named milestones).
Topology: my-first-lab.clab.yml; map: my-first-lab.clab.yml.annotations.json.
```

Then commit and push:

```
$ git add my-first-lab
$ git commit -m "my-first-lab: topology and map"
$ git push
$ git status -sb
```

**Expected result** `git status -sb` reports `## main...origin/main` — clean, nothing left to push. On GitHub, `my-first-lab/` now holds `README.md`, `my-first-lab.clab.yml`, `my-first-lab.clab.yml.annotations.json` and `latest/`. The checkout must stay clean and pushed like this before the manager's next **Save progress**, or that save is refused.

```
B7 - owner terminal, lab VM (as clabllm)

$ cd /home/clabllm/labs/my-network-labs && mkdir -p my-first-lab && cp /srv/containerlab-node-manager/projects/my-first-lab/my-first-lab.clab.yml* my-first-lab/

$ cat > my-first-lab/README.md # 5-line description written above

$ git add my-first-lab

$ git commit -m "my-first-lab: topology and map"
[main 127bfe8] my-first-lab: topology and map
3 files changed, 65 insertions(+)
create mode 100644 my-first-lab/README.md
create mode 100644 my-first-lab/my-first-lab.clab.yml
create mode 100644 my-first-lab/my-first-lab.clab.yml.annotations.json

$ git push
To https://github.com/pruger-dev/my-network-labs.git
a6121e5..127bfe8 main -> main

$ git status -sb
## main...origin/main
```

Figure B.7 — Publishing the topology and map from the lab VM.

## B8 Check GitHub

**Where to run it** On your computer.

**Action** Open `my-network-labs` on GitHub.

**Expected result** The repository shows a `my-first-lab` folder holding the topology file, its map, the README you wrote, and the `latest/` folder your save uploaded. Back in the manager, the same tree is one click away: **Progress > Saved versions > Browse the repository...** opens the repository browser on the lab's own folder and lists `README.md`, `my-first-lab.clab.yml`, `my-first-lab.clab.yml.annotations.json` and `latest/` side by side — the in-product view of exactly what GitHub holds.

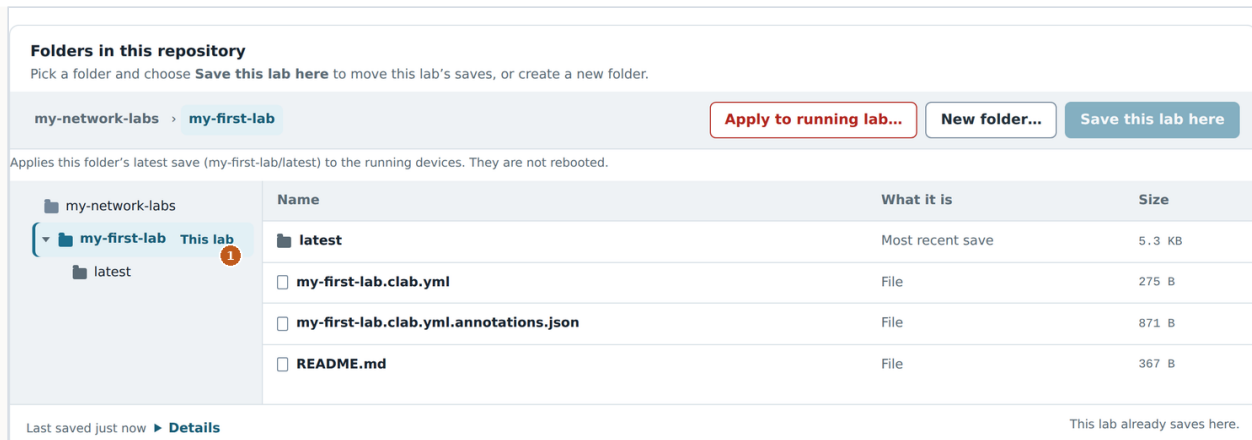


Figure B.8 — Browsing the repository from the Progress tab.

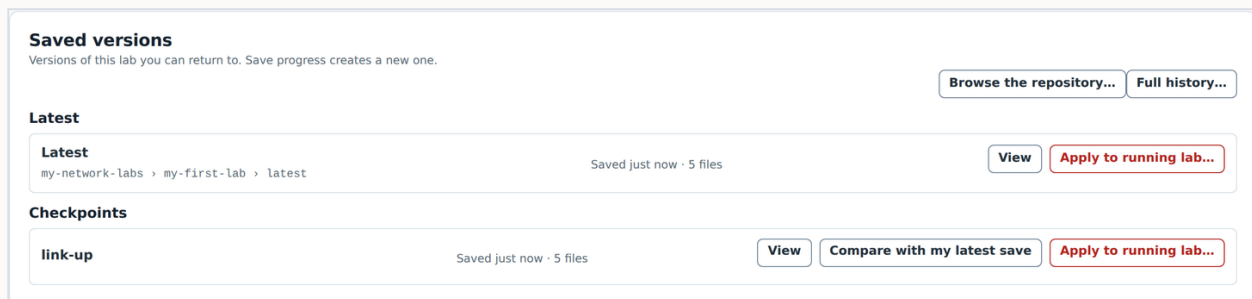
### B9 Change, save, and checkpoint again

**Action** On `r2`'s CLI, add a `Loopback0` addressed `10.255.0.2/32`, the same way as in Scenario A, then `write memory`. Click **Save progress**.

**Expected result** This time **Review before uploading** lists only `r2.cfg` changed and `r2.eoscfg` changed — `r1` did not change. Click **Upload these changes**; the status reads **Saved to Git just now**.

**Action** Open the **Save progress** menu and choose **Create checkpoint...**, name it `link-up`, and upload that review too.

**Expected result** **Saved versions** lists **Latest** (`my-network-labs > my-first-lab > latest`) and, under **Checkpoints**, `link-up`. On GitHub, `my-first-lab/latest/r2.cfg` contains the new `Loopback0`, `my-first-lab/checkpoints/link-up/` exists, and there is still no nested `latest/latest` — the two saves simply added two more commits ("Save my-first-lab progress") above "my-first-lab: topology and map".

Figure B.9 — Checkpoint `link-up` alongside `Latest`.

### B10 Prove it: destroy, remove, and rebuild from Git

**Action** From **Lab actions** ▾, choose **Destroy lab...**. The review, titled **Destroy my-first-lab?**, warns the same way as in Scenario A; confirm with **Destroy lab**.

**Expected result** The lab header reads **Stopped** — the devices are gone.

**Action** From **Lab actions ▼**, choose **Remove from this manager...** and confirm with **Remove lab**. Its checkbox **Don't offer this lab for import again** can stay ticked or not — either is fine here. This step only removes the lab from this manager; it never touches your repository or the topology file you published in B7.

**Where to run it** On the lab VM, as your ordinary account. Clone your repository into a trusted lab folder, since the manager's deploy browser only shows folders it trusts:

```
$ git clone https://github.com/pruger-dev/my-network-labs.git /srv/containerlab-node-manager/projects/my-network-labs
```

**Expected result** Cloning into '/srv/containerlab-node-manager/projects/my-network-labs'...

**Action** Back in the manager, from Home click **Choose a file on the lab VM...**, open /srv/containerlab-node-manager/projects → my-network-labs → my-first-lab (the folder inside the clone) → click ◇ **my-first-lab.clab.yml**. In the **Topology file** dialog, check that **File location on the VM** reads /srv/containerlab-node-manager/projects/my-network-labs/my-first-lab/my-first-lab.clab.yml — it must contain my-network-labs. The original my-first-lab folder is still listed at the top level and looks the same; picking that one instead would deploy the old copy, not the one you just rebuilt from Git. Click **Deploy lab**, then confirm the **Start my-first-lab?** review with **Start lab**.

**Expected result** Both devices read **Ready** after about 45 seconds, and the Topology tab shows the same map as before — the annotations file came from Git along with the topology.

**Action** Open the **Progress** tab and **Connect a repository by URL** again, with the same HTTPS URL and folder my-first-lab.

**Expected result** The card simply reads "my-first-lab saves to my-network-labs > my-first-lab > latest/" again — the checkout registered in B6 is reused, with no special message.

**Action** Under **Saved versions**, find **Latest** and click **Apply to running lab...** The review, titled **Replace running configuration**, lists both devices with their differences; confirm with **Replace configurations**.

**Expected result** Within about 12 seconds each device reports "Configuration replaced and verified against the saved desired state." On r1's CLI, ping 10.0.0.2 succeeds, and show ip interface brief on r2 lists Loopback0 10.255.0.2/32 — the lab is rebuilt entirely from what Git kept.

### Topology file

1

File location on the VM

/srv/containerlab-node-manager/projects/my-network-labs/my-first-lab/my-first-lab.clab.yml

Topology (YAML)

```

name: my-first-lab
topology:
  nodes:
    r1:
      kind: arista_ceos
      image: n24l/ceos:4.35.0F
      mgmt-ipv4: 172.20.20.21
    r2:
      kind: arista_ceos
      image: n24l/ceos:4.35.0F
      mgmt-ipv4: 172.20.20.22
  links:
    - endpoints: [ "r1:eth1", "r2:eth1" ]

```

2

Deploy lab adds this lab to My labs and starts its devices on the VM. Add without starting keeps it in My labs only. Existing files can't be edited as text here — use Edit visually, or edit them on the VM.

Preview topology

Edit visually...

Add to My labs without starting

Deploy lab

Figure B.10a — The Topology file dialog, confirming the path runs through my-network-labs.

### Configuration replaced

• Configuration replaced

Configuration replaced on 2 devices.

• Replaced and verified r1 EOS

► Details

• Replaced and verified r2 EOS

► Details

► Details

Figure B.10b — Latest applied; both devices replaced and verified.

## Checklist

- [ ] my-first-lab drawn, imaged correctly and deployed
- [ ] Link addressed and pinging
- [ ] my-network-labs created on GitHub, private, with a first commit
- [ ] Repository connected, first **Save progress** uploaded
- [ ] Topology and map published from the lab VM; checkout clean and pushed
- [ ] A second change saved, and checkpoint link-up created
- [ ] Lab destroyed and removed, then redeployed from the folder inside the cloned repository
- [ ] Repository reconnected and **Latest** applied; the link pings and r2's loopback is back

## Reference

### What is saved where

| What it holds                                | Where it lives                                      | Created by                                                                    | Survives a device restart?                                                                   | Survives a Destroy lab?                                                                                |
|----------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| A drawn but unsaved topology                 | This browser's storage                              | The lab builder, as you edit                                                  | Not applicable — no lab is deployed yet                                                      | Not applicable                                                                                         |
| The topology file and its map                | A lab folder on the lab VM                          | <b>Save to the VM...</b> / <b>Deploy lab</b> / <b>Write a new topology...</b> | Yes                                                                                          | Yes — Destroy removes the running devices and the generated lab folder, not the original topology file |
| A device's running configuration             | The device itself                                   | Commands you type; <code>write memory</code> on EOS                           | Only if written to startup                                                                   | No — Destroy removes the containers                                                                    |
| A local progress save ("Saved on the VM")    | The registered checkout on the lab VM               | <b>Save progress</b> , then <b>Not now — keep it on the VM</b>                | Yes                                                                                          | Yes                                                                                                    |
| An uploaded <b>Latest</b>                    | Your online repository                              | <b>Upload these changes</b> / <b>Upload now</b>                               | Yes                                                                                          | Yes                                                                                                    |
| A named checkpoint                           | Your online repository (and locally until uploaded) | <b>Create checkpoint...</b>                                                   | Yes                                                                                          | Yes                                                                                                    |
| The result of <b>Apply to running lab...</b> | The running devices                                 | <b>Apply to running lab...</b> , confirmed with <b>Replace configurations</b> | Yes — on EOS the manager runs <code>write memory</code> for you once the change is confirmed | No — Destroy removes the containers; redeploy and apply again                                          |

### Stop, Destroy, Remove

**Stop devices** keeps the containers so you can start them again quickly; nothing configured is lost, but it is also not saved anywhere by stopping. **Destroy lab...** removes the running devices (and, on this manager, the generated lab folder with them) — configuration changes you never saved are lost, but your saved progress, checkpoints and backups all remain, since they were never stored on the devices. **Remove from this manager...** only forgets the lab

in this manager; it never deletes anything on the lab VM or in your repository, and a lab it removed can always be added back by deploying its topology file again.

## Troubleshooting

| Symptom                                                                                  | Next step                                                                                                                                                  | Ask your instructor if...                                                 |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| A device stays on <b>Starting</b> for several minutes                                    | Wait — cEOS and similar network operating systems take up to a couple of minutes to boot and answer a login                                                | It never reaches <b>Ready</b>                                             |
| A device shows <b>Needs credentials</b>                                                  | The device is up, but the manager could not log in with the configured credentials                                                                         | You do not know the intended login for this lab                           |
| A save reads <b>Saved on this VM — not uploaded</b>                                      | Open it under <b>Recent saves</b> and click <b>Review and upload...</b> , then <b>Upload these changes</b>                                                 | The upload keeps failing after several tries                              |
| An upload or sync is rejected                                                            | Check you are connecting your own copy of the repository, not a read-only one, and that the acknowledgement was ticked                                     | The rejection names a permission or repository problem you cannot resolve |
| No <b>Apply to running lab...</b> button, or a device is listed as skipped in the review | Read the reason shown next to that device — a platform this manager cannot restore yet, or a saved version made before this device's platform supported it | The reason is not one of these                                            |
| Deploy fails naming an image the VM does not have                                        | Open the lab with <b>Edit visually...</b> , correct the device's <b>Image</b> field, save, and deploy again                                                | The image you were told to use still is not found                         |

## Next session in 60 seconds

- Open the manager at `http://192.168.132.132:8081`.
- Open your lab from Home and click **Start lab** if it is not already running.
- Wait for every device to read **Ready**.
- Check the **Progress** tab: does it still say **Saved to Git**, or is something waiting under **Recent saves**?
- Pick up where you left off, or apply a saved version to jump to a specific point.
- End the session with **Save progress**, and **Upload these changes** if you want it kept online.